Modern Physics – I

Photoelectric effect

Light: 1675 Newton : Corpuscular theory

1800 : YDSE
Diffraction
Interference
Polarization
→ light is behaving like wave

1905 : Photoelectric experiment → light is particle

Einstein stated that light has dual nature → Wave, Particle

Particle nature of light / EM wave:

→ Particle of light is known as 'Photon' or 'Quanta'.
→ Photon has speed 3×10^8 m/s in vacuum.
→ Photon is electrically neutral. Hence, it does not create or respond to electric and magnetic fields.
→ Photon has rest mass zero
→ Photon can be created (bulb, sun) or destroyed (absorbing body)
→ No. of photons is not conserved.
→ When photon is absorbed, then momentum conservation and energy conservation are valid.
→ Photon has momentum : $P = \frac{h}{\lambda}$

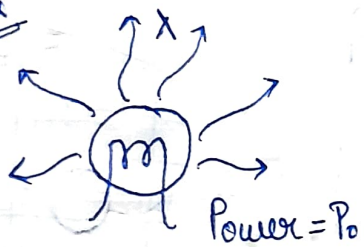
$h \rightarrow$ planck's constant
$h = 6.626\times10^{-34}$ J-s $= 4.14\times10^{-15}$ eV-sec

$1eV = 1.6\times10^{-19}$ J

$\vec{P} \parallel \vec{C}$

→ Photon has energy $E = hf = \frac{hc}{\lambda}$ $\quad E = PC$ $\quad P = \frac{E}{C}$

Ex*



LASER type bulb (i.e emits only one wavelength)

Find number of photons emitted per second.

$E = Pt = P_0 t$

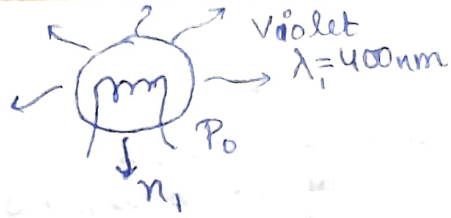
In 1 sec $E = P_0$

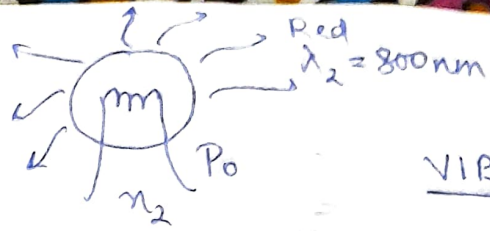
Energy of Photons : $E = n\left(\frac{hc}{\lambda}\right)$

So, $P_0 = n\frac{hc}{\lambda}$

* $n = \frac{P_0\lambda}{hc}$

Ex





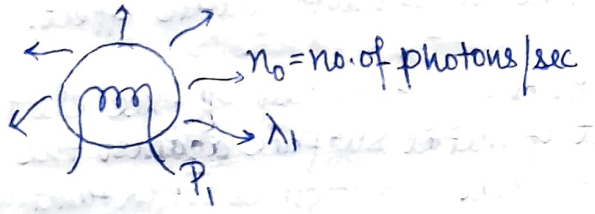
VIBGYOR → $\lambda\uparrow$

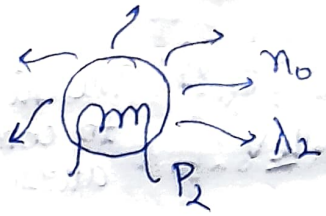
$$n_1 = \frac{P_0 \lambda_1}{hc} \qquad n_2 = \frac{P_0 \lambda_2}{hc}$$

$$\star \boxed{\frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2}}$$

$$\frac{n_1}{n_2} = \frac{400}{800} = \frac{1}{2} \Rightarrow \boxed{n_2 = 2n_1}$$

Ex

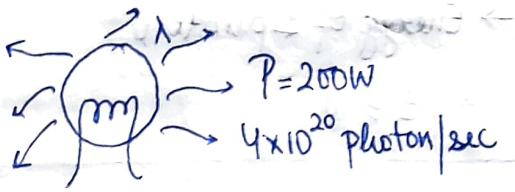




$$P_1 = \frac{n_0 hc}{\lambda_1} \qquad P_2 = \frac{n_0 hc}{\lambda_2}$$

$$\star \boxed{\frac{P_1}{P_2} = \frac{\lambda_2}{\lambda_1}}$$

Q



Find λ

$$P = \frac{nhc}{\lambda} \Rightarrow \lambda = \frac{nhc}{P}$$

$$= \frac{4 \times 10^{20} \times 6.626 \times 10^{-34} \times 3 \times 10^8}{200}$$

$\boxed{1nm = 10^{-9}m}$
$\boxed{1\text{Å} = 10^{-10}m}$

$$\lambda = 39.756 \times 10^{-8} m$$
$$= 397.56 nm$$
$$= 3975.6 \text{Å}$$

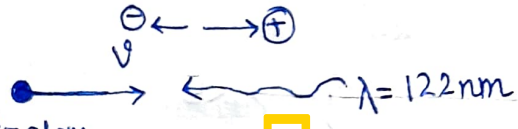
Q



Find no. of photons emitted/sec.

$$n = \frac{P\lambda}{hc} = \frac{100 \times 5 \times 10^{-7}}{6.626 \times 10^{-34} \times 3 \times 10^8} = 25.15 \times 10^{19} \text{ photon/s}$$

Q



$m = 1.67 \times 10^{-27} kg$

H atom absorbs the photon and stops. Find v.

Momentum of H-atom alone of photon is not conserved but momentum of system is conserved.

M/C $\Rightarrow P_i = P_f$

$$mv + \left\{-\frac{h}{\lambda}\right\} = 0 + 0$$

$$v = \frac{h}{m\lambda}$$

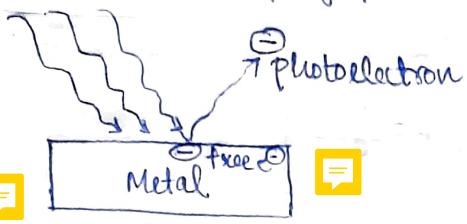
$$v = \frac{6.626 \times 10^{-34}}{1.67 \times 10^{-27} \times 122 \times 10^{-9}}$$

$$v = 3.25 m/s$$

Photoelectric effect:

→ When light or EM wave of sufficiently high frequency is irradiated on a metal surface, free $e^{\ominus}$s on metal surface may absorb them and may come out of metal opposing the attractive force by metal. The emitted electron is known as photoelectron and any current due to these photoelectrons is known as photocurrent. This entire phenomenon is known as photoelectric effect.

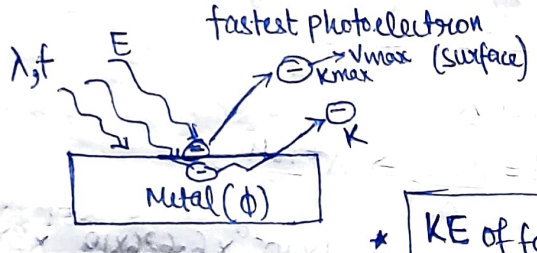
Light or em wave of light.



Work function $\{\phi, W\}$: The minimum energy required by free electron to come out of metal surface against the attraction of metal is known as work function.

Ex Cs ⇒ W = 1.9eV
smallest WF in the Periodic Table

Einstein's Equation:



E → Energy of 1 photon

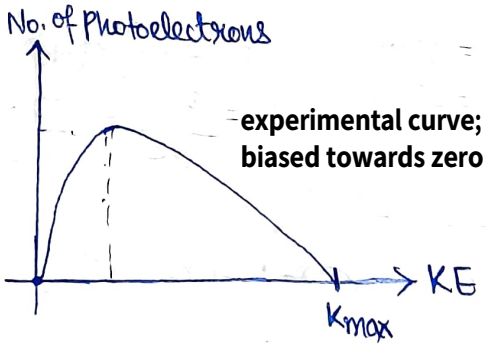
* $E \geq \phi$

* KE of fastest photoelectron $= K_{max} = E - \phi = \frac{hc}{\lambda} - \phi = hf - \phi$

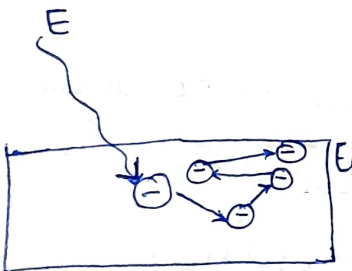
Other photoelectron ⇒ $K < E - \phi$

$\therefore$ $0 \leq K \leq E - \phi$
$0 \leq K \leq hf - \phi$
$0 \leq K \leq \frac{hc}{\lambda} - \phi$

Einstein's eqn.



experimental curve; the maxima is slightly biased towards zero KE



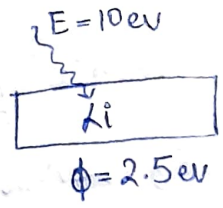
This free $e^{\ominus}$ absorbed photon but due to its internal collisions its final energy is less than ϕ. So this $e^{\ominus}$ cannot come out of metal surface.

* No. of photoelectrons << No. of photons incident

Quantum efficiency = Efficiency of P.E.

$= \eta = \frac{\text{no. of photoelectrons emitted}}{\text{no. of photons incident}} \times 100$

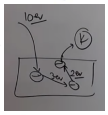
Ex

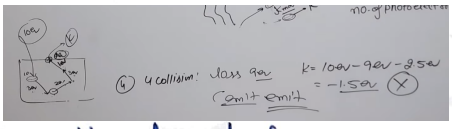


① No collision ⇒ K = 10 − 2.5 = 7.5 eV = Kmax

② 1 collision, loss = 3 eV ⇒ K = 10 − 3 − 2.5 = 4.5 eV

③ 2 collision, loss = 5 eV ⇒ K = 10 − 5 − 2.5 = 2.5 eV





Threshold Frequency $\{f_{Th}\}$: It is the least frequency which can just emit an electron out of a metal.

$$E \geq \phi \Rightarrow E_{min} = \phi = hf_{Th} = \phi$$

$$\therefore \boxed{f_{Th} = \frac{\phi}{h}}$$

Also, $K_{max} = E - \phi = hf - hf_{Th}$

$$\boxed{K_{max} = h(f - f_{Th})}$$

Q Find f_{Th} of Cs ⇒ $\phi = 1.9$ eV

$\phi = hf_{Th}$

$1.9\ eV = \frac{6.626 \times 10^{-34}}{1.6 \times 10^{-19}}\ eV\text{-}s\ f_{Th}$

$1.9 = 4.14 \times 10^{-15}\ f_{Th}$

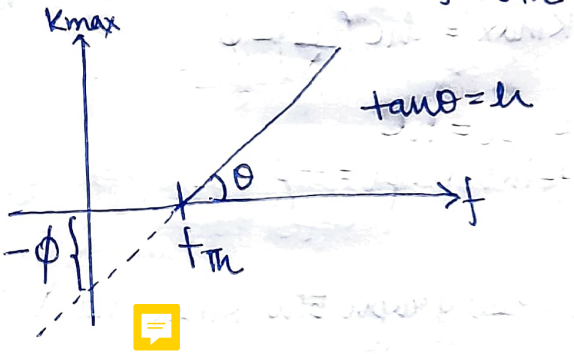
$f_{Th} = \frac{1.9 \times 10^{15}}{4.14} = 4.58 \times 10^{14}\ Hz$

Q Draw graph b/w Kmax of emitted photo-electron and frequency of light incident.

$K_{max} = hf - \phi$ Slope = h

$K_{max} = h(f - f_{Th})$ Y-intercept = $-\phi$

For PEE ⇒ $f \geq f_{Th}$



Note:

$$K_{max} = \frac{hc}{\lambda} - \phi$$

$$\boxed{K_{max} = \frac{12420\ eV\text{-}Å}{\lambda(Å)} - \phi(eV)}$$

↳ This will give kmax in eV.

$$\boxed{1\ eV = 1.6 \times 10^{-19}\ J}$$

$hc = 6.626 \times 10^{-34} \times 3 \times 10^{8}$ SI units

$hc = 4.14 \times 10^{-15}\ eV\text{-}sec \times 3 \times 10^{18}\ Å/sec$

$hc = 12.42 \times 10^{3}\ eV\text{-}Å$

$hc = 12420\ eV\,Å$

$hc = 1242\ eV\text{-}nm$

Threshold Wavelength: Maxm wavelength of a light which can just emit an electron from metal surface is known as threshold wavelength.

$E \geq \phi \Rightarrow E_{min} = \phi = \frac{hc}{\lambda_{Th}} = \phi \Rightarrow$ $\boxed{\lambda_{Th} = \frac{hc}{\phi}}$ $\boxed{\lambda_{Th} = \frac{12420\ eV\text{-}\mathring{A}}{\phi\ (eV)}}$

$\boxed{\lambda_{Th} f_{Th} = C}$

For PEE, $\lambda \leq \lambda_{Th}$

Q $\boxed{\phi = 2ev}$ Find its ① λ_{Th} ② f_{Th}

① $\lambda_{Th} = \frac{hc}{\phi} = \frac{12420\ ev\text{-}\mathring{A}}{2\ ev} = 6210\mathring{A}$

② $\lambda_{Th} f_{Th} = C \Rightarrow f_{Th} = \frac{C}{\lambda_{Th}}$

$f_{Th} = \frac{3 \times 10^8\ m/s}{6210 \times 10^{-10}\ m} = 4.83 \times 10^{14}\ Hz$

For PEE, $f \geq 4.83 \times 10^{14}\ Hz$

$\lambda \leq 6210\mathring{A}$

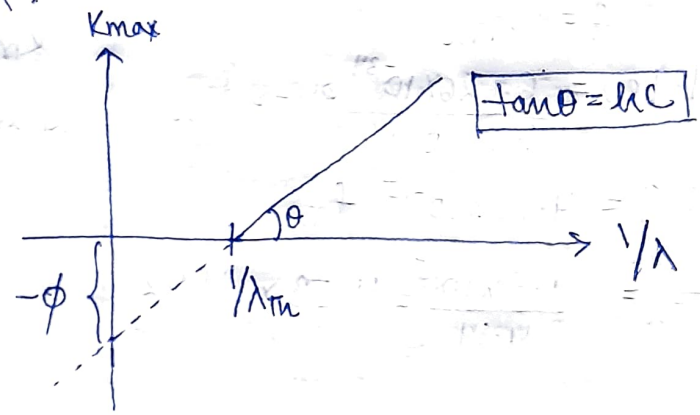
Q Draw graph b/w Kmax and $1/\lambda$.

$K_{max} = \frac{hc}{\lambda} - \phi$

$K_{max} = hc\left(\frac{1}{\lambda}\right) - \phi$

Slope $= hc$

y-intercept $= -\phi$



Q Draw graph b/w Kmax and λ.

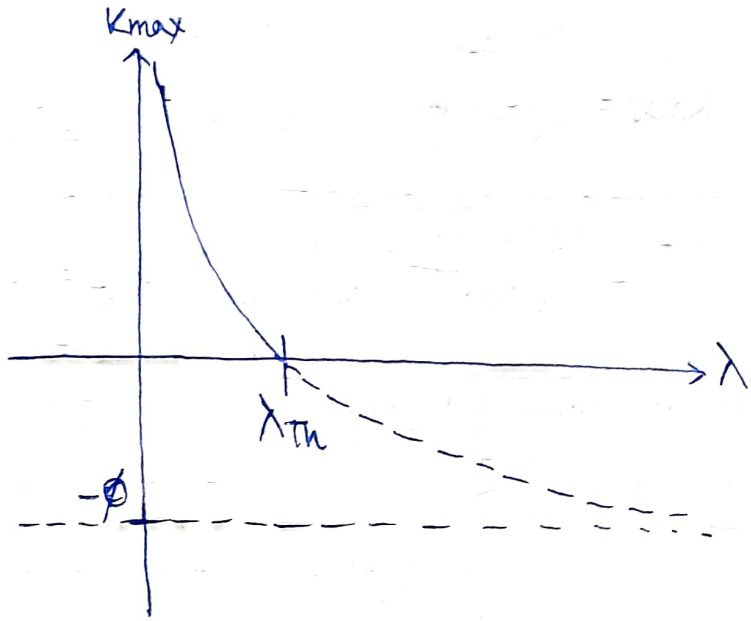
$K_{max} = \frac{hc}{\lambda} - \phi$

$(K_{max} + \phi)\lambda = hc$

It is of the type

$\underbrace{(Y + Y_0)X = C}$

Rectangular Hyperbola with shifted origin at $-Y_0$



Alternate form of Einstein's equation:

$$K_{max} = E - \phi = hf - \phi = \frac{hc}{\lambda} - \phi$$

$$K_{max} = \frac{hc}{\lambda} - \frac{hc}{\lambda_{th}} = hc\left\{\frac{1}{\lambda} - \frac{1}{\lambda_{th}}\right\}$$

$$K_{max} = hf - hf_{th} = h(f - f_{th})$$

Q Given a metal with $\lambda_{th} = 6210\,Å$. Find

① Work function of metal: $\phi = \frac{hc}{\lambda_{th}} = \frac{12420\ eV\text{-}Å}{6210\,Å} = 2\,eV$

② KE of fastest emitted $e^{\ominus}$ if light of wavelength $7000\,Å$ is incident.
Here, $\lambda > \lambda_{th} \Rightarrow$ No PEE $\Rightarrow$ No photoelectrons emitted.

③ KE of fastest emitted $e^{\ominus}$ if light of wavelength $3105\,Å$ is incident.

$$K_{max} = \frac{hc}{\lambda} - \phi = \frac{12420\ eVÅ}{3105\,Å} - 2eV = 4eV - 2eV = 2eV$$

Q Given a metal with $f_{th} = 5 \times 10^{14}\,Hz$. Find

① Work function of metal: $\phi = hf_{th} = 4.14 \times 10^{-15}\ eV\,\cancel{s}/\cancel{sec} \times 5 \times 10^{14}\,\cancel{Hz} = 2.07\,eV$

② If light of $f = 10^{15}\,Hz$ is incident on it, find K_{max} of emitted photoelectron.

$$K_{max} = hf - \phi = 4.14 \times 10^{-15} \times 10^{15} - 2.07 = 2.07\,eV$$

③ For the same frequency, find speed of fastest emitted photoelectron.

$$K_{max} = \frac{1}{2}mv^2 = 2.07\,eV \Rightarrow v^2 = \frac{2 \times 2.07 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}\ m^2/s^2$$

Q An EM wave of equation $E = 50\,\frac{V}{m}\{\sin(5\times10^{15}t) + \sin(8\times10^{15}t)\}$ is incident on a metal surface of $\phi = 2eV$. Find KE of fastest photoelectron.

$E = 50\{\sin(5\times10^{15}t) + \sin(8\times10^{15}t)\}$

This wave is a superposition of two waves at $x = 0$.

$$E_1 = 50 \sin 5\times10^{15}\left(t - \frac{x}{v}\right)$$

$$E_2 = 50 \sin 8\times10^{15}\left(t - \frac{x}{v}\right)$$

$\omega_1 = 5\times10^{15}$ $\quad$ $\omega_2 = 8\times10^{15}$

$\omega_2 > \omega_1 \Rightarrow \omega_2$ will eject fastest photoelectron

$$f = \frac{8\times10^{15}}{2\pi}$$

$$K_{max} = hf - \phi$$

$$= 4.14\times10^{-15}\ eV\text{-}\cancel{s} \times \frac{8\times10^{15}\,\cancel{Hz}}{2\times3.14} - 2eV$$

$$= \frac{4.14\times4}{3.14} - 2$$

$$= 5.27 - 2$$

$$= 3.27\,eV$$

Q When light of $\lambda_1 = 5400\,Å$ strikes the metal the speed of fastest photoelectron is v. When light of $\lambda_2 = 3500\,Å$ strikes the metal, the speed of fastest photoelectron is twice of the initial value. Find ϕ of metal.

$$\frac{1}{2}mv^2 = K = \frac{hc}{\lambda_1} - \phi \quad -(1)$$

$$\frac{1}{2}m(2v)^2 = 4K = \frac{hc}{\lambda_2} - \phi \quad -(2)$$

So, $\phi = \frac{1}{3}\left\{\frac{4hc}{\lambda_1} - \frac{hc}{\lambda_2}\right\}$

$$\phi = \frac{1}{3}\left\{\frac{4\times 12420}{5400} - \frac{12420}{3500}\right\}$$

$$= \frac{1}{3}\{9.2 - 3.54\} = 1.88\,eV$$

Divide (2) by (1)

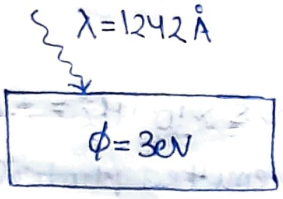
$$\rightarrow 4 = \frac{\frac{hc}{\lambda_2} - \phi}{\frac{hc}{\lambda_1} - \phi}$$

$$= \frac{4hc}{\lambda_1} - 4\phi = \frac{hc}{\lambda_2} - \phi$$

$$= 3\phi = \frac{4hc}{\lambda_1} - \frac{hc}{\lambda_2}$$

$$\phi = \frac{1}{3}\left\{\frac{4hc}{\lambda_1} - \frac{hc}{\lambda_2}\right\}$$

** Q



An $e^{\ominus}$ absorbs the photon and looses 10% of its available energy. Find its K.E. when it emits out of the metal after 5 collisions in the lattice.

$e^{\ominus}$ looses 10% of the available energy after every collision $\Rightarrow$ 90% of the available energy is left every time.

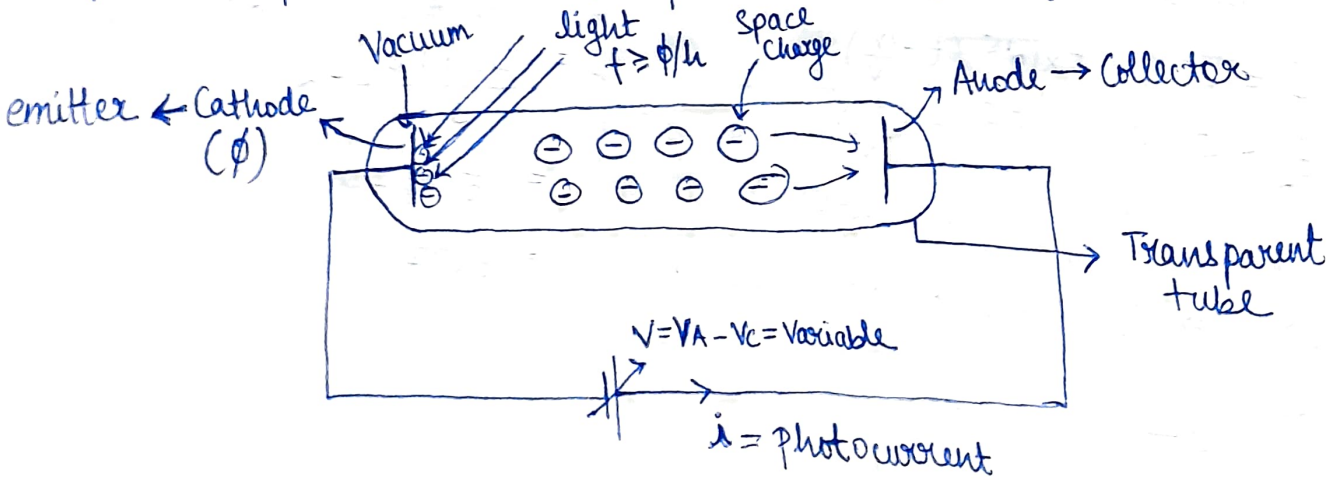
So,

10 eV	$\rightarrow$	$10\times\frac{90}{100}$	$\rightarrow$	$10\times\frac{90}{100}\times\frac{90}{100}$	$\rightarrow$	$10\times\left(\frac{90}{100}\right)^3$	$\rightarrow$	$10\times\left(\frac{90}{100}\right)^4$	$\rightarrow$	$10\left(\frac{90}{100}\right)^5$
		1st collision		2nd Collision		3rd		4th		5th

$$K = E - \phi = 10\left(\frac{90}{100}\right)^5 - 3 = 5.9049 - 3 = 2.9049\,eV$$

Also if $10\left(\frac{90}{100}\right)^n < 3 \Rightarrow e^{\ominus}$ cant come out {n = no. of collisions}

Experiment of Photoelectric effect {Henrick Hertz (1887)}:



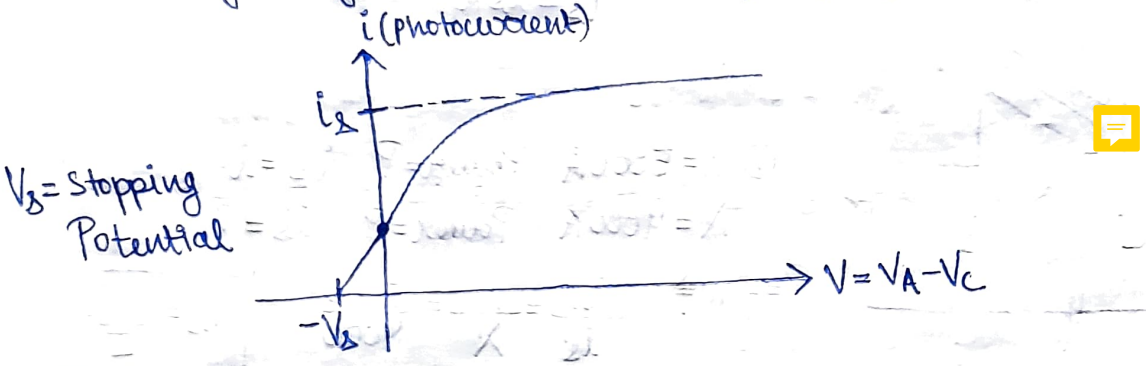
* If $V=0$, then photoelectrons with sufficiently high K.E. will reach anode and contribute in photocurrent. But photoelectrons with small KE may get repelled by space charge and hence may return back to the cathode and hence will not contribute in photocurrent.

* Accelerating Voltage $\Rightarrow V_{anode} > V_{cathode} \Rightarrow V_A - V_C = +ve$

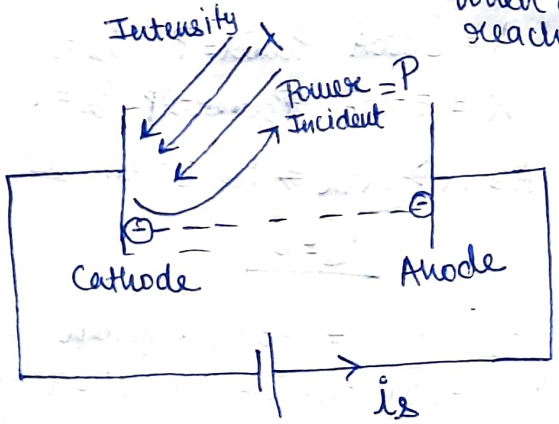
As $V_A - V_C \uparrow \Rightarrow E \uparrow \Rightarrow$ Force on $e^{\ominus} \uparrow$

* When all the photoelectrons emitted by cathode reach anode, we get the maximum value of current called saturation current (i_s).

* Retarding Voltage $\Rightarrow V_{Anode} < V_{cathode} \Rightarrow V_A - V_C = -ve$.



Saturation Current (i_s): It is the maximum value of photocurrent when all the photoelectrons emitted by cathode reach anode.



In 1 sec,

$E = P \times 1 = \frac{hc}{\lambda} \times n_p$

$\boxed{n_p = \frac{P\lambda}{hc}} \Rightarrow$ No. of photons incident.

Quantum efficiency $= \eta$

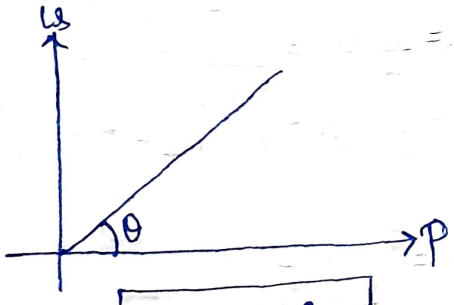
$\eta = \frac{n_e}{n_p} \times 100$

$\Rightarrow n_e = \frac{\eta n_p}{100} = \frac{\eta P \lambda}{100hc} =$ No. of photoelectrons emitted per sec.

$i_s = \frac{q}{t} = \frac{\frac{\eta P \lambda e}{100hc} \times t}{t} = \frac{\eta P \lambda e}{100hc}$

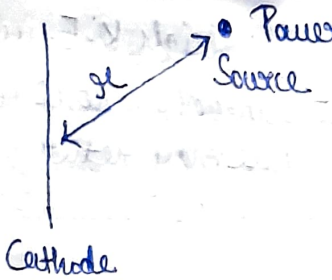
$\therefore \boxed{i_s = \frac{\eta P \lambda e}{100hc}}$

If λ = fix $\Rightarrow i_s \propto P$



$\boxed{\tan\theta = \frac{\eta \lambda e}{100hc}}$

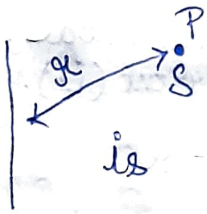
Note:

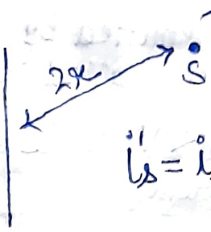


$$P = IA = \frac{P_{out} \cdot A}{4\pi r^2}$$

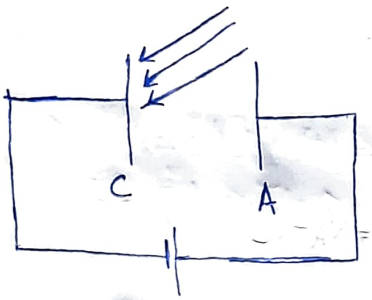
$$\therefore \boxed{\text{Power} \propto \frac{1}{r^2} \Rightarrow i_s \propto \frac{1}{r^2}}$$

Ex





Q



① $\lambda = 5000\text{Å}$ Power $= P$ $i_s = i$

$\lambda' = 4000\text{Å}$ Power $= P$ $i_s' = ?$

$P = fix \Rightarrow \frac{i_s}{i_s'} = \frac{\lambda}{\lambda'} = \frac{5000}{4000}$ $\boxed{i_s' = \frac{4}{5} i}$

② $\lambda = 5000\text{Å}$ $n_p = n$ $i_s = i$

$\lambda' = 4000\text{Å}$ $n_p = n$ $i_s' = ?$

$$\frac{i_s'}{i_s} = \frac{P'\lambda'}{P\lambda} = \frac{\frac{nhc}{\lambda'} \times \lambda'}{\frac{nhc}{\lambda} \times \lambda} = 1$$

$\therefore \boxed{i_s' = i_s = i}$

③ $\lambda_{th} = 4000\text{Å}$

$\lambda = 3000\text{Å}$ Power $= P$ $i_s = i$

$\lambda' = 5000\text{Å}$ Power $= 2P$ $i_s' = ?$

As $\lambda' > \lambda_{th} \Rightarrow$ No $e^\ominus$ emitted

$\therefore \boxed{i_s' = 0}$

Q A photocell is working in saturation mode with $i = 4.8\mu A$ when radiation of $\lambda = 3000\text{Å}$ and $P = 1\text{mW}$ is incident. When radiation of $\lambda = 1650\text{Å}$ and $P' = 5\text{mW}$ is incident, it is observed that max^m speed of $e^\ominus$ becomes twice. Find ① ϕ of cell ② Quantum efficiency ③ i_s in second case.

$\lambda = 3000\text{Å}$ $P = 1\text{mW}$ $i = 4.8\mu A$

$\lambda' = 1650\text{Å}$ $P' = 5\text{mW}$ $i_s' = ?$

① $K = \frac{1}{2}mv^2 = \frac{hc}{\lambda} - \phi$

$4K = \frac{1}{2}m(2v)^2 = \frac{hc}{\lambda'} - \phi$

$\Rightarrow 4 = \frac{\frac{hc}{\lambda} - \phi}{\frac{hc}{\lambda'} - \phi}$

$\Rightarrow \frac{4hc}{\lambda'} - 4\phi = \frac{hc}{\lambda} - \phi$

$\Rightarrow \phi = \frac{1}{3}\left\{\frac{4hc}{\lambda'} - \frac{hc}{\lambda'}\right\}$

$= \frac{1}{3}\left\{\frac{4 \times 12420}{3000} - \frac{12420}{1650}\right\} ev$

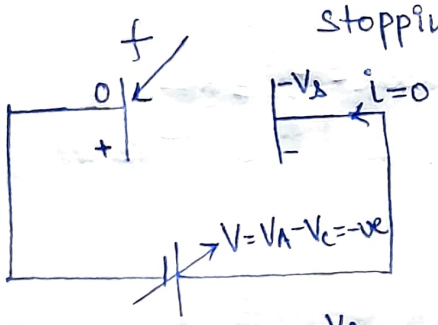
$= \phi = 3.01 ev$

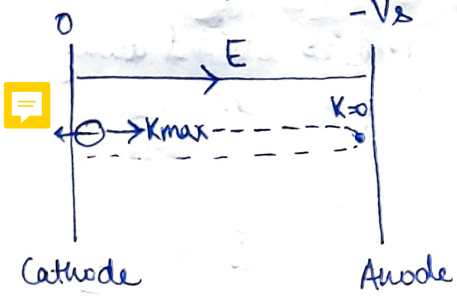
(2) Quantum efficiency $\Rightarrow$ $i_s = \frac{P\lambda\eta e}{100hc} \Rightarrow \eta = \frac{i\,100hc}{P\lambda e}$.

$$\eta = \frac{4.8\times10^{-6}\times100\times6.626\times10^{-34}\times3\times10^{8}}{10^{-3}\times3000\times10^{-10}\times1.6\times10^{-19}} = 1.99\%.$$

(3) i_s' → $\frac{i'}{i} = \frac{P'\lambda'}{P\lambda} \Rightarrow i' = \frac{P'\lambda'}{P\lambda} i = \frac{5mW\times1650\text{Å}}{1mW\times3000\text{Å}} \times 4.8\mu A = 13.2\mu A$

Stopping Potential: Min^m negative potential of Anode wrt cathode at which photocurrent just becomes zero and no $e^{\ominus}$ emitted by cathode reaches anode, is known as stopping potential.





Using TMEC:

$K_{max} + (-e)(0) = 0 + (-e)(-V_s)$

$K_{max} = eV_s$

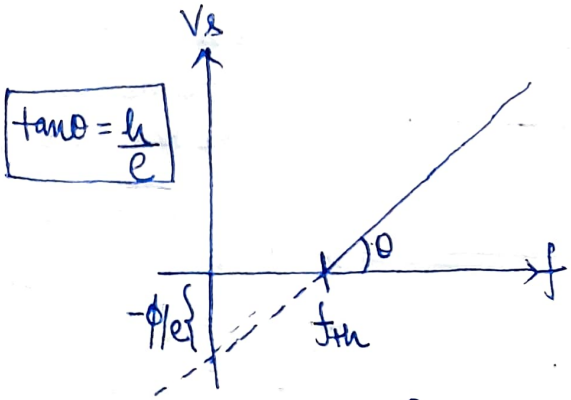
$K_{max} = 5eV \quad V_s = \frac{5eV}{e} = 5Volt$

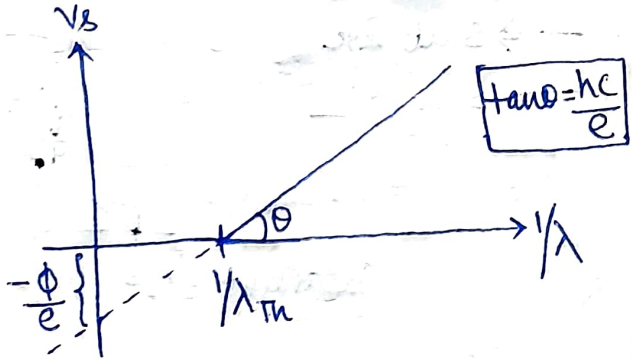
$K_{max} = 6eV \quad V_s = \frac{6eV}{e} = 6Volt$

$\star$ $\boxed{V_s = \frac{K_{max}}{e}}$

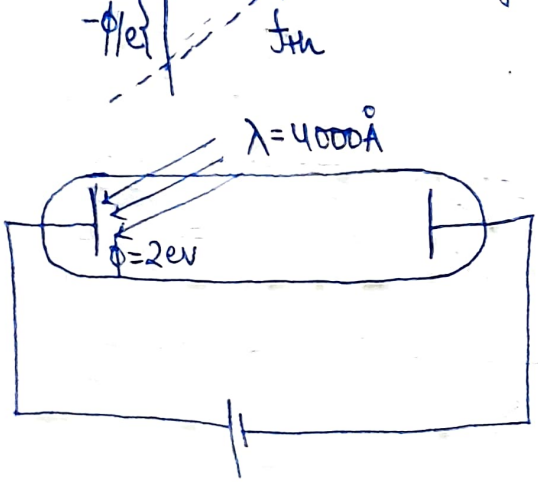
Also, $K_{max} = eV_s = hf - \phi = \frac{hc}{\lambda} - \phi$

$\boxed{V_s = \frac{hf}{e} - \frac{\phi}{e}}$ $\boxed{V_s = \frac{hc}{\lambda e} - \frac{\phi}{e}}$





Q



Find V_s

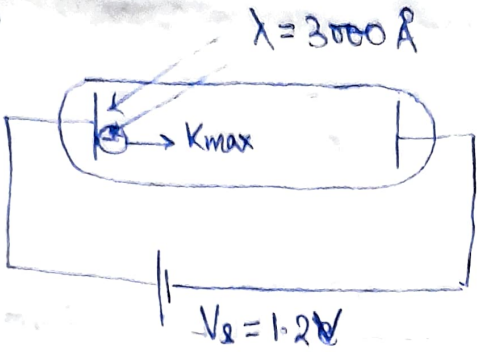
$K_{max} = \frac{hc}{\lambda} - \phi = \frac{12420\,eV\text{Å}}{4000\,\text{Å}} - 2eV$

$K_{max} = 1.105\,eV$

$K_{max} = eV_s$

$1.105\,eV = e\,V_s$ $\Rightarrow$ $\boxed{V_s = 1.105\,V}$

Q



Find ϕ of the cathode plate.

$K_{max} = eV_s = 1.2eV = \frac{hc}{\lambda} - \phi$

$1.2eV = \frac{12420\ eV\text{Å}}{3000\ \text{Å}} - \phi$

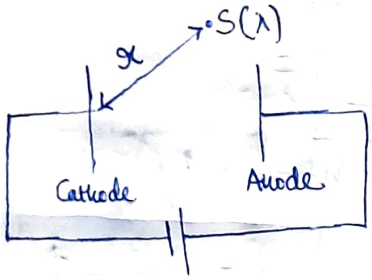
$\phi = \left(\frac{1242}{300} - 1.2\right) eV = 2.94\ eV$

Parameters affecting stopping potential / KEmax

$V_s = \frac{hc}{\lambda e} - \frac{\phi}{e}$

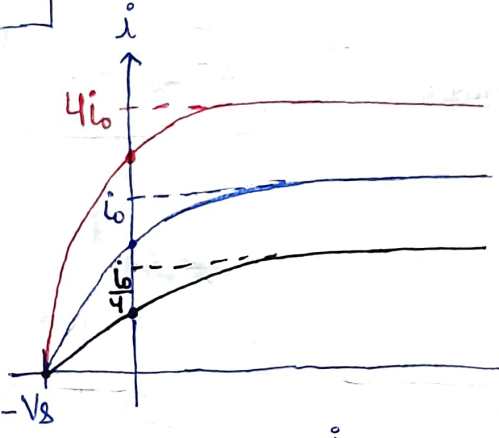
- depends on energy of 1 photon
- doesn't depend on number of photons
- Energetic photon ⇒ E↑, f↑, λ↓ ⇒ V_s↑
- If ϕ↓ ⇒ V_s↑

Q



Draw graph if source is shifted away and separation with plate becomes 2x and x/2.

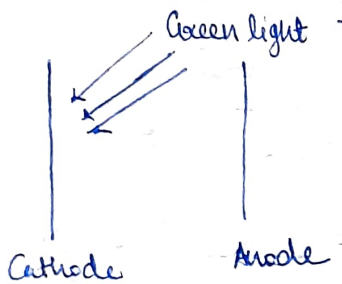
— ⇒ S at x
— ⇒ S at 2x
— ⇒ S at x/2



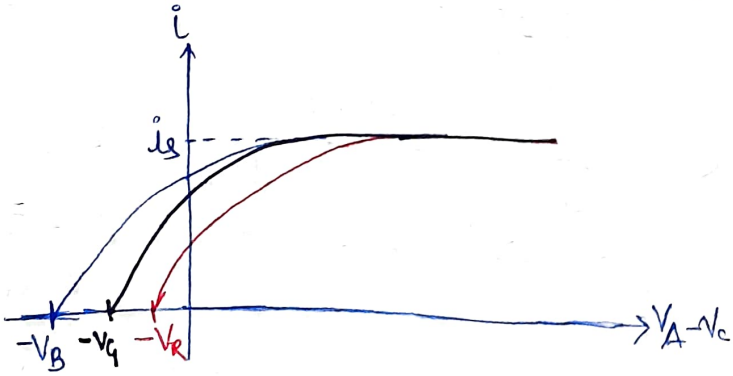
$i_s \propto \frac{1}{x^2}$

As λ of light remains same V_s remains same because energy of each photons remains same.

Q



Keeping No. of photons to be same, if red and blue light is incident. Draw the graph

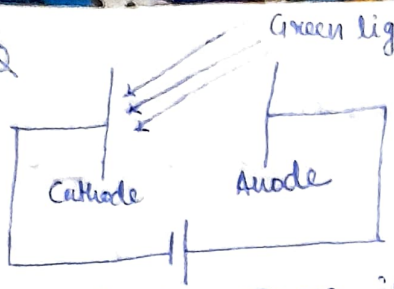


$V_s \propto \frac{1}{\lambda}$ VIBGYOR → λ↑

$\lambda_B < \lambda_G < \lambda_R$

Also as no. of photons are same, no. of photoelectrons will also be same ⇒ i_s same.

Q

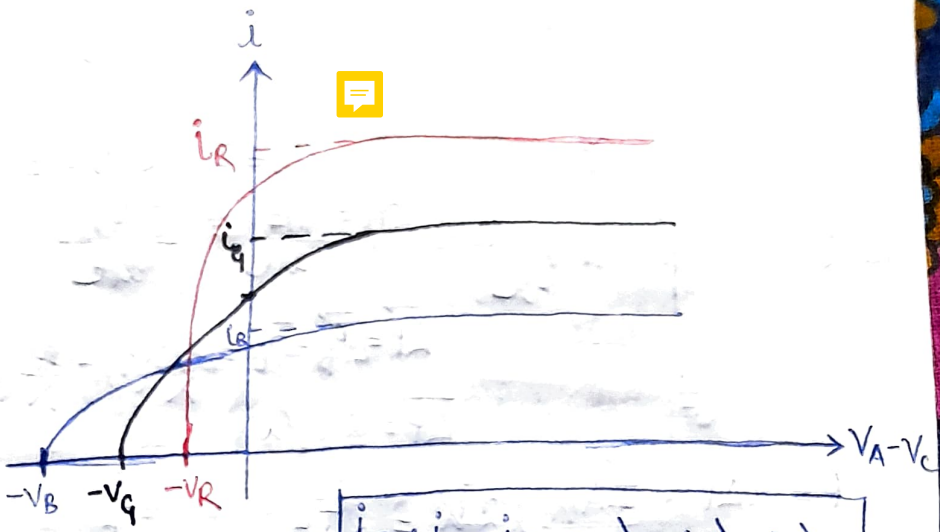


Considering same power, if red and blue light are incident. Draw i v/s $V_A - V_C$ graph.

$n = \frac{P\lambda}{hc}$ P is same $\Rightarrow$ $n \propto \lambda$

$V_s \propto \frac{1}{\lambda}$

Also, if $n\uparrow$, $\Rightarrow$ more photo electrons $\rightarrow$ more saturation current

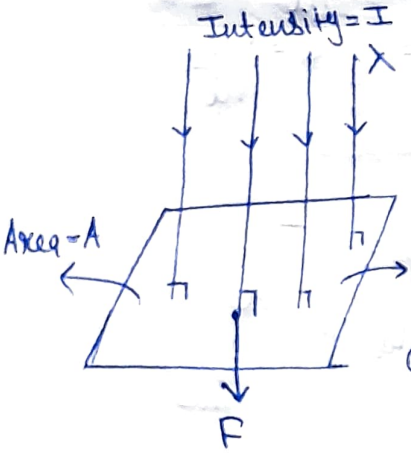


$i_R : i_g : i_B = \lambda_R : \lambda_g : \lambda_B$

VIBGYOR $\rightarrow$ $\lambda\uparrow$

Radiation Force:

① Normal incidence – Perfectly absorbing surface:



Absorbtive power $= a = 1$

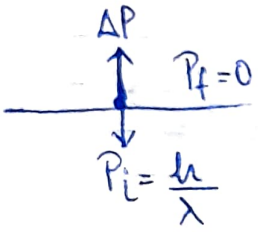
$a = \frac{\text{no. of photons absorbed}}{\text{no. of photons incident}}$

Power incident $= P = \vec{I} \cdot \vec{A}$

$P = IA\cos 0^\circ = IA$

No. of photons incident/sec:

$n = \frac{P\lambda}{hc} = \frac{IA\lambda}{hc}$



$\vec{\Delta P} = \vec{P_f} - \vec{P_i}$

$= 0 - \frac{h}{\lambda}(\downarrow)$

$\vec{\Delta P} = \frac{h}{\lambda}(\uparrow)$

$F = \frac{\Delta P}{t} = N\frac{\frac{h}{\lambda}}{t} = \frac{nh}{\lambda} = n\Delta P$

N = No. of photons incident

n = No. of photons incident/sec

$F = n\Delta P = \frac{IA\lambda}{hc} \times \frac{h}{\lambda} = \frac{IA}{c}$

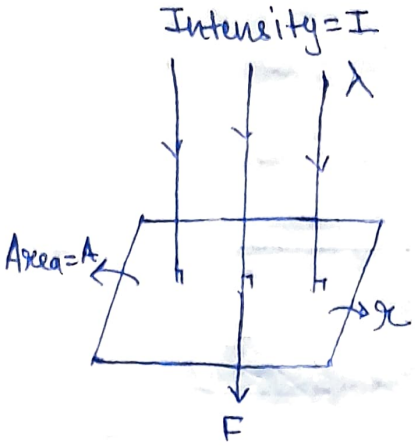
$\text{Pressure} = \frac{F_\perp}{A_\perp} = \frac{IA}{cA} = \frac{I}{c}$

★ $F = \frac{IA}{c} = \frac{\text{Power}}{c}$

★ $\text{Pressure} = \frac{I}{c}$

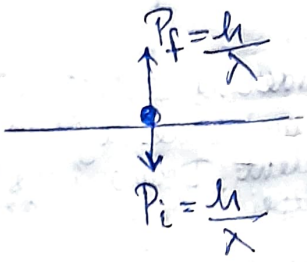
② Normal Incidence – Perfectly reflecting surface:

→ absorbtive power = 0
→ coeff of reflection = $r = 1 = \frac{n_R}{n_i}$



$$\text{Power} = IA\cos 0 = IA$$

$$n = \frac{P\lambda}{hc} = \frac{IA\lambda}{hc}$$



$$\vec{\Delta P} = \vec{P_f} - \vec{P_i} = \frac{h}{\lambda}(\uparrow) - \frac{h}{\lambda}(\downarrow)$$

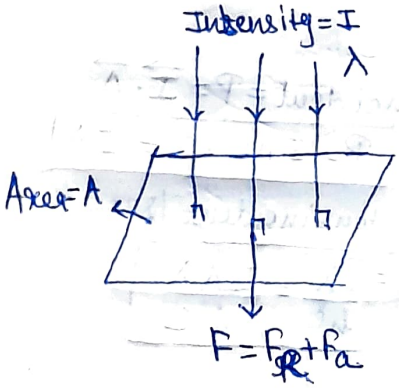
$$\vec{\Delta P} = \frac{2h}{\lambda}(\uparrow)$$

$$F = n\Delta P = \frac{IA\lambda}{hc} \times \frac{2h}{\lambda} = \frac{2IA}{c}$$

$$\text{Pressure} = \frac{F}{A} = \frac{2I}{c}$$

$$\star \boxed{F = \frac{2IA}{c} = \frac{2 \times \text{Power}}{c}}$$

③ Normal Incidence – Partially reflecting surface: $r = \frac{n_R}{n_i}$



$$\text{Power} = IA \qquad a = 1 - r = \frac{n_a}{n_i}$$

$$n_i = \frac{P\lambda}{hc} = \frac{IA\lambda}{hc}$$

$$\text{No. of photons reflected/time} = n_R = r\frac{IA\lambda}{hc}$$

$$\text{No. of photons absorbed/time} = n_a = (1-r)\frac{IA\lambda}{hc}$$

$$\Delta P_R = \frac{2h}{\lambda} \qquad \Delta P_a = \frac{h}{\lambda}$$

$$F = F_R + F_a = n_R\Delta P_R + n_a\Delta P_A$$

$$= \frac{rIA\lambda}{hc} \times \frac{2h}{\lambda} + \frac{(1-r)IA\lambda}{hc} \times \frac{h}{\lambda}$$

$$\boxed{F = \frac{(1+r)IA}{c} = \frac{(1+r)\,\text{Power}}{c}}$$

$$\boxed{\text{Pressure} = \frac{F}{A} = \frac{(1+r)I}{c}}$$

a) If $r = 0 \Rightarrow F = \frac{IA}{c} \Rightarrow$ case ①

b) If $r = 1 \Rightarrow F = \frac{2IA}{c} \Rightarrow$ case ②

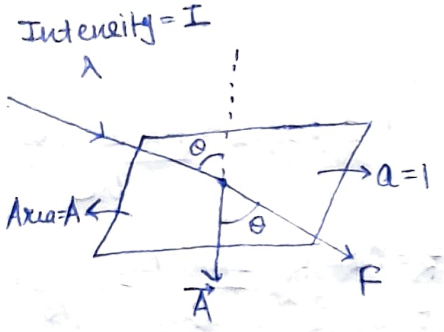
a) If $r = 0 \Rightarrow P = \frac{I}{c}$

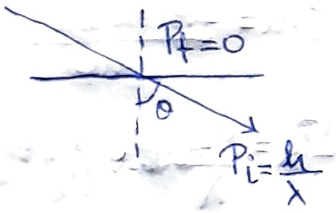
b) If $r = 1 \Rightarrow P = \frac{2I}{c}$

④ Oblique Incidence – Perfectly absorbing surface:

$$\text{Power} = IA\cos\theta$$

$$n = \frac{P\lambda}{hc} = \frac{IA\lambda\cos\theta}{hc}$$





$$\Delta\vec{P} = \vec{P_f} - \vec{P_i}$$

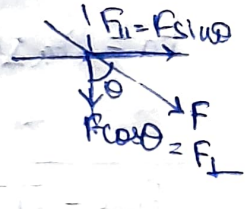
$$= 0 - \frac{h}{\lambda}\left(\text{unit vector at } \theta\right)$$

$$\Delta\vec{P} = \frac{h}{\lambda}\left(\text{unit vector at } \theta\right)$$

$$F = n\Delta P = \frac{IA\lambda\cos\theta}{hc} \times \frac{h}{\lambda} = \frac{IA\cos\theta}{c}$$

$$\star \boxed{F = \frac{IA\cos\theta}{c}}$$

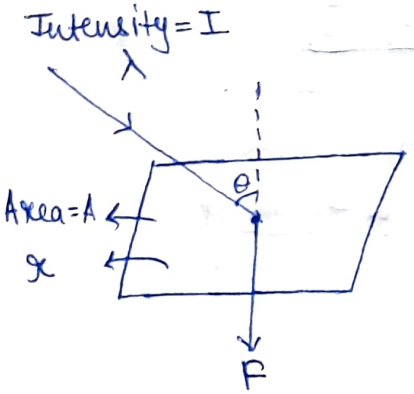
$$\star \boxed{\text{Pressure} = \frac{F_\perp}{A} = \frac{F\cos\theta}{A} = \frac{I\cos^2\theta}{c}}$$



$$\star \boxed{\text{Shear Stress} = \frac{F_\parallel}{A} = \frac{F\sin\theta}{A} = \frac{I\cos\theta\sin\theta}{c}}$$

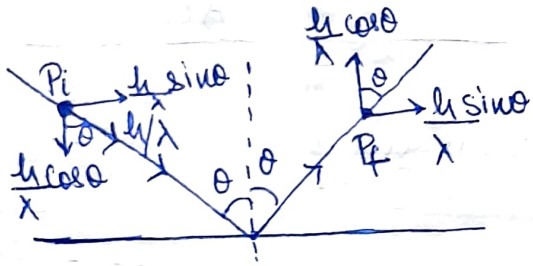
$$\frac{n\theta}{A} = \frac{I\sin 2\theta}{2c} \quad \theta = 45° \Rightarrow \left(\text{Shear stress}\right)_{max} = \frac{I}{2c}$$

⑤ Oblique Incidence – Perfectly reflecting surface:



$$\text{Power} = IA\cos\theta$$

$$n = \frac{P\lambda}{hc} = \frac{IA\cos\theta\lambda}{hc}$$



$$\Delta\vec{P} = \vec{P_f} - \vec{P_i}$$

$$= \left(\frac{h}{\lambda}\sin\theta\,\hat{i} + \frac{h}{\lambda}\cos\theta\,\hat{j}\right) - \left(\frac{h}{\lambda}\sin\theta\,\hat{i} - \frac{h}{\lambda}\cos\theta\,\hat{j}\right)$$

$$\Delta\vec{P} = \frac{2h}{\lambda}\cos\theta\,\hat{j}$$

$$F = n\Delta P = \frac{IA\lambda\cos\theta}{hc} \times \frac{2h}{\lambda}\cos\theta$$

$$\star \boxed{F = \frac{2IA\cos^2\theta}{c}}$$

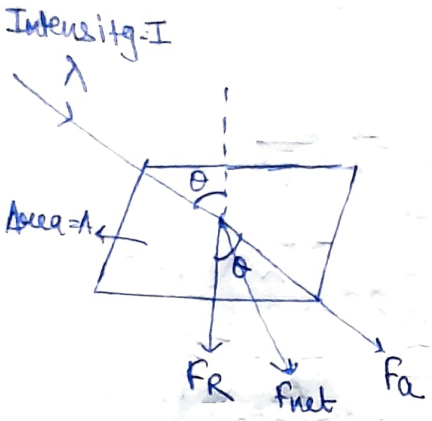
$$\text{Pressure} = \frac{F}{A} = \frac{2IA\cos^2\theta}{cA}$$

$$\star \boxed{\text{Pressure} = \frac{2I\cos^2\theta}{c}}$$

⑥ Oblique Incidence – Partially reflecting surface: $r = \frac{n_R}{n_i}$, $a = 1 - r = \frac{n_a}{n_i}$

Power $= IA\cos\theta$

$n_i = \frac{P\lambda}{hc} = \frac{IA\lambda\cos\theta}{hc}$



$n_R = r n_i = \frac{IA\lambda r\cos\theta}{hc}$ $\quad n_a = (1-r)n_i = \frac{(1-r)IA\lambda\cos\theta}{hc}$

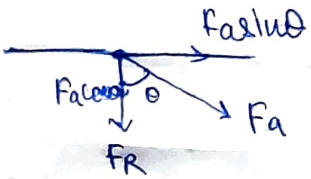
$\Delta P_R = \frac{2h}{\lambda}\cos\theta(1)$ $\quad F_R = n_R\Delta P_R = \frac{IA\lambda r\cos\theta}{hc} \times \frac{2h}{\lambda}\cos\theta$

$\Delta P_a = \frac{h}{\lambda}(1)$ $\quad \star$ $\boxed{F_R = \frac{2IAr\cos^2\theta}{c}}$

$F_a = n_a\Delta P_a = \frac{(1-r)IA\lambda\cos\theta}{hc} \times \frac{h}{\lambda} = \frac{(1-r)IA\cos\theta}{c}$ $\quad \star$ $\boxed{F_a = \frac{(1-r)IA\cos\theta}{c}}$

$\boxed{F_{net} = \sqrt{F_R^2 + F_a + 2F_RF_a\cos\theta}}$

$\text{Pressure} = \frac{F_R + F_a\cos\theta}{A} = \frac{2IAr\cos^2\theta + (1-r)IA\cos^2\theta}{cA}$



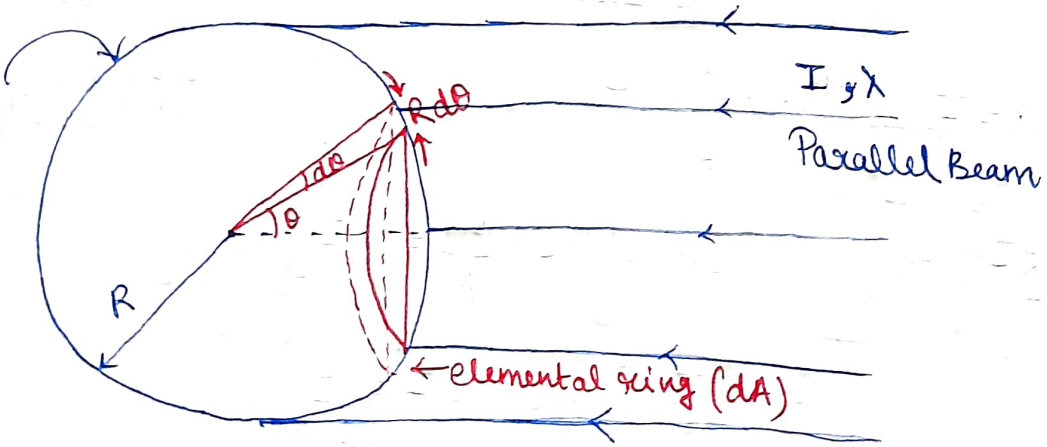
$\star$ $\boxed{\text{Pressure} = \frac{(1+r)I\cos^2\theta}{c}}$

$\star$ $\boxed{\text{Shear Stress} = \frac{F_a\sin\theta}{A} = \frac{(1-r)IA\cos\theta\sin\theta}{Ac} = \frac{(1-r)I\sin 2\theta}{2c}}$

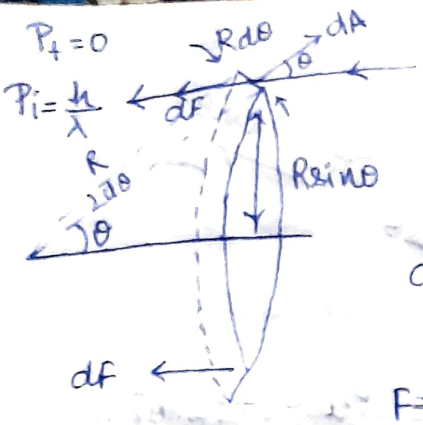
$\star\star\star$ Q Find force due to radiation on the sphere.

Sol

Perfectly absorbing surface $a=1$



$dA = 2\pi R\sin\theta \cdot R d\theta = 2\pi R^2\sin\theta d\theta$



$$\frac{\text{No. of photons incident}}{\text{Time}} = \frac{P\lambda}{hc} = \frac{IA\cos\theta\,\lambda}{hc}$$

$$n = \frac{I\,2\pi R^2 \sin\theta\cos\theta\,\lambda\,d\theta}{hc}$$

$$dF = n\Delta P = \frac{2\pi R^2 I \sin\theta\cos\theta\,d\theta}{hc} \times \frac{h}{\lambda}$$

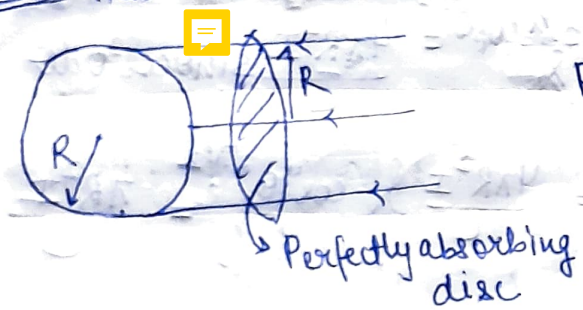
$$\vec{\Delta P} = 0 - \frac{h}{\lambda}(\leftarrow)$$

$$\vec{\Delta P} = \frac{h}{\lambda}(\rightarrow)$$

$$F = \int dF = \frac{2\pi R^2 I}{c}\int_0^{\pi/2} \sin\theta\cos\theta\,d\theta = \frac{2\pi R^2 I}{c}\left\{\frac{\sin^2\theta}{2}\right\}_0^{\pi/2}$$

$$F = \frac{\pi R^2 I}{c}\left\{\sin^2\frac{\pi}{2} - \sin^2 0\right\} \quad \therefore \boxed{F = \frac{\pi R^2 I}{c}}$$

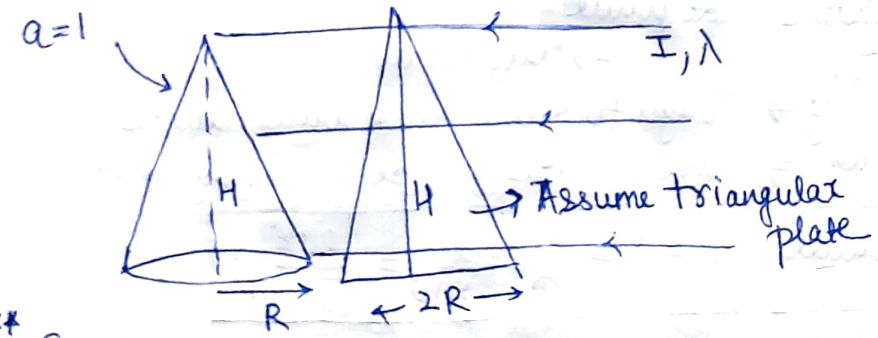
Method2: Trick



$$F = \frac{IA}{c} = \frac{I\pi R^2}{c}$$

trick is valid only for perfectly absorbing bodies

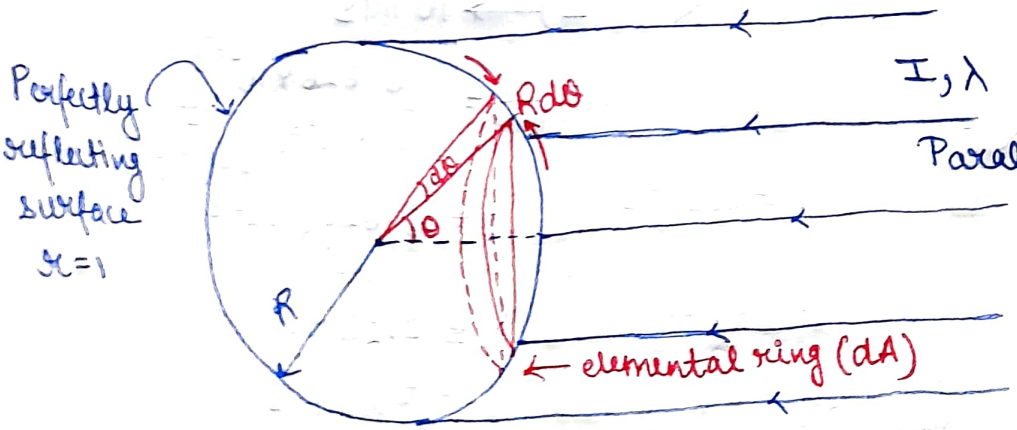
Q Find force due to radiation on cone



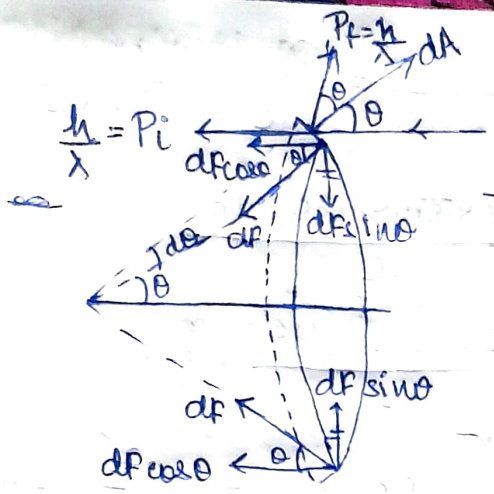
$$F = \frac{IA}{c} = \frac{I}{c} \times \frac{1}{2} \times 2R \times H$$

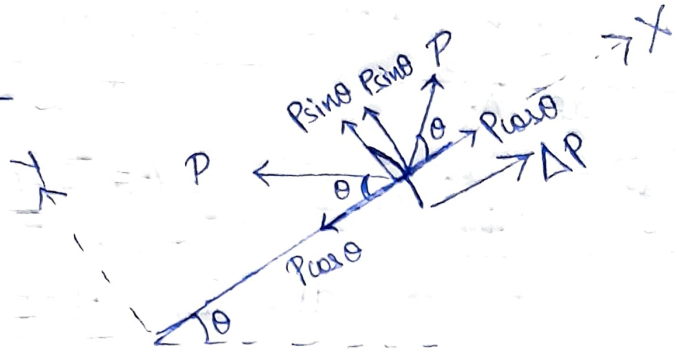
$$\boxed{F = \frac{IRH}{c}}$$

** Q Find force due to radiation on the sphere.



$$dA = 2\pi R^2 \sin\theta\,d\theta$$





$$\vec{\Delta P} = \vec{P_f} - \vec{P_i} = (P\cos\theta\,\hat{i} + P\sin\theta\,\hat{j}) - (-P\cos\theta\,\hat{i} + P\sin\theta\,\hat{j})$$

$$\vec{\Delta P} = 2P\cos\theta\,\hat{i} = \frac{2h}{\lambda}\cos\theta\,\hat{i}$$

$$n = \frac{P\lambda}{hc} = \frac{I\,2\pi R^2 \sin\theta\cos\theta d\theta\,\lambda}{hc}$$

$$dF = n\Delta P = \frac{I\,2\pi R^2 \sin\theta\cos\theta d\theta}{hc}\lambda \times \frac{2h}{\lambda}\cos\theta = \frac{4\pi R^2 I}{c}\sin\theta\cos^2\theta d\theta$$

$$F = \int dF\cos\theta = \frac{4\pi R^2 I}{c}\int_0^{\pi/2}\cos^3\theta\sin\theta d\theta = \frac{4\pi R^2 I}{c}\left\{-\frac{\cos^4\theta}{4}\right\}_0^{\pi/2} = \frac{\pi R^2 I}{c}$$

Matter Wave / De-Broglie's Hypothesis:

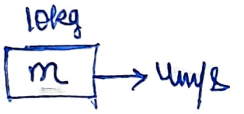
→ Particle / matter can behave like wave
↳ Matter wave / de-broglie wave.

→ Heavy object / Slowly moving ⇒ Tough to show their wave nature
Light object / fast moving ⇒ Wave nature can be shown by experiments.

$m \rightarrow v$

De-broglie wavelength = $\lambda = \frac{h}{P} = \frac{h}{mv} = \frac{h}{\sqrt{2mKE}}$

Q



$$\lambda = \frac{h}{mv} = \frac{6.626\times10^{-34}}{10\times4}$$

$$\lambda = \frac{6.626\times10^{-33}}{4}$$

$$\lambda = 1.6\times10^{-33}\ m \quad (\text{very small})$$

Q



$$\lambda = \frac{h}{mv} = \frac{6.626\times10^{-34}}{9.1\times10^{-31}\times10^{5}}$$

$$\lambda = \frac{6.626}{9.1}\times10^{-8}$$

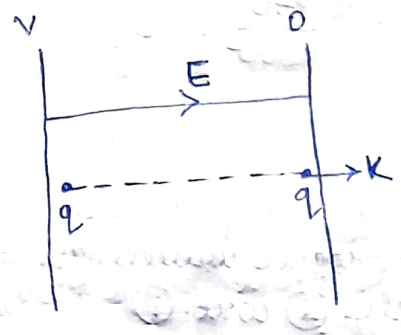
$$\lambda = 7.28\times10^{-9}\ m$$

$$\lambda = 7.28\ nm$$

De-broglie wavelength of $e^{\ominus}$:

$\ominus \rightarrow v$

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mKE}} = \frac{h}{\sqrt{2mqV}}$$



Using TMEC

$0 + qV = K + q(0)$

$\star \quad K = qV$

V = Potential through which charge is accelerated

For electron:

$$\lambda = \frac{h}{\sqrt{2meV}} = \sqrt{\frac{150}{V}}\,\text{Å} = \frac{12.27}{\sqrt{V}}\,\text{Å}$$

Q Find de-broglie wavelength of $e^{\ominus}$ which has been accelerated by a P.d. of 6 volts.

$$\lambda = \sqrt{\frac{150}{V}}\,\text{Å} = \sqrt{\frac{150}{6}}\,\text{Å} = \sqrt{25}\,\text{Å} = 5\,\text{Å}$$

Q Find ratio of debroglie wavelength of proton and α-particle having

proton $\Rightarrow$ $m = 1\,amu$, $q = +e$

α-particle $\Rightarrow$ $m = 4\,amu$, $q = +2e$

① Same speed: $\lambda = \frac{h}{mv}$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{h/m_p v_p}{h/m_\alpha v_\alpha} = \frac{m_\alpha}{m_p} = \frac{4}{1}$$

② Same KE: $\lambda = \frac{h}{\sqrt{2mKE}}$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{h/\sqrt{2m_p K_p}}{h/\sqrt{2m_\alpha K_\alpha}} = \sqrt{\frac{m_\alpha}{m_p}} = \frac{2}{1}$$

③ Accelerated through same V: $\lambda = \frac{h}{\sqrt{2mqV}}$

$$\frac{\lambda_p}{\lambda_\alpha} = \frac{h/\sqrt{2m_p q_p V_p}}{h/\sqrt{2m_\alpha q_\alpha V_\alpha}} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}} = \sqrt{\frac{8}{1}} = 2\sqrt{2}$$

De-broglie wavelength of gaseous molecule:

$n \longrightarrow v$ $\quad P = mv = \sqrt{2mKE} = \sqrt{2m\frac{3}{2}RT} = \sqrt{3mRT}$

$$KE = \frac{3}{2}RT$$

R → Boltzman Const

$$\star \quad \lambda = \frac{h}{mv} = \frac{h}{\sqrt{3mRT}}$$

$$\lambda = \frac{h}{\sqrt{3\frac{M}{N_A}\cdot\frac{R}{N_A}T}}$$

$$\star \quad \lambda = \frac{hN_A}{\sqrt{3MRT}}$$

M = molar mass in kg

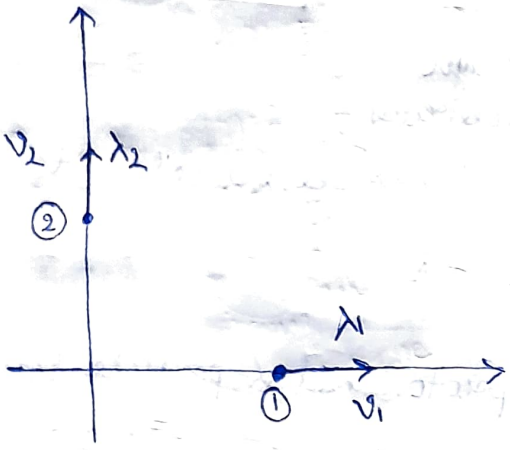
R = Gas Const

T = Temp in Kelvin

Q Find ratio of de-broglie wavelength of H_2 gas at 27°C and He gas at 127°C.

$$\frac{\lambda_{H_2}}{\lambda_{He}} = \frac{h N_A / \sqrt{3 M_{H_2} R T_{H_2}}}{h N_A / \sqrt{3 M_{He} R T_{He}}} = \sqrt{\frac{M_{He} T_{He}}{M_{H_2} T_{H_2}}} = \sqrt{\frac{4}{2} \times \frac{400}{300}} = \sqrt{\frac{8}{3}}$$

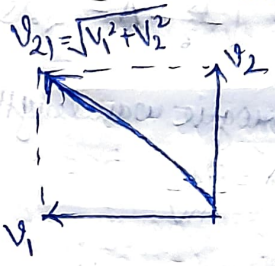
Q



Given two identical particles. Find de-broglie wavelength of particle ② wrt ①

$$\lambda_1 = \frac{h}{m v_1} \Rightarrow v_1 = \frac{h}{m \lambda_1}$$

$$\lambda_2 = \frac{h}{m v_2} \Rightarrow v_2 = \frac{h}{m \lambda_2}$$



$$\lambda_{21} = \frac{h}{m v_{21}}$$

$$\lambda_{21} = \frac{h}{m\sqrt{v_1^2 + v_2^2}} = \frac{h}{m\sqrt{\frac{h^2}{m^2\lambda_1^2} + \frac{h^2}{m^2\lambda_2^2}}} = \frac{\lambda_1 \lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}$$

$$\therefore \boxed{\lambda_{21} = \frac{\lambda_1 \lambda_2}{\sqrt{\lambda_1^2 + \lambda_2^2}}}$$

thus, De-Broglie wavelength depends upon the frame of reference